



ANDERSON SERANGOON JUNIOR COLLEGE

2021 JC 2 PRELIMINARY EXAMINATION

H1 CHEMISTRY

8873/01

Paper 1 Multiple Choice

22 September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name and class on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

Multiple Choice Answer Sheet

Write your name, class and NRIC / FIN number, **including** the reference letter.

Shade the NRIC / FIN number.

Exam Title: JC2 Preliminary Exam

Exam Details: H1 Chemistry / Paper 1

Date: 22/09/2021

This document consists of **14** printed pages and **2** blank pages.

- 1 *Use of the Data Booklet is relevant to this question.*

Neon exists as three isotopes, ^{20}Ne , ^{21}Ne and ^{22}Ne .

How many neutrons are present in a neon atom of the most abundant isotope?

- A 10 B 11 C 12 D 20

- 2 Carbon disulfide, CS_2 , is a poisonous, volatile liquid used in the manufacture of viscose rayon and cellophane. It reacts with nitrogen monoxide, NO , to form a yellow solid and equal volumes of two gaseous products.

What are the products of the reaction?

- A S, CO , N_2
B S, CO , NO_2
C S, CO_2 , N_2
D S, CO_2 , NO_2

- 3 *Use of the Data Booklet is relevant to this question.*

How many hydrogen atoms are present in 4.5 g of butanone?

[L = Avogadro constant]

- A $72L$ B L C $\frac{L}{2}$ D $\frac{L}{16}$

- 4 A 10 cm^3 gaseous sample of butanethiol, $\text{CH}_3(\text{CH}_2)_3\text{SH}$, was sparked in 100 cm^3 of oxygen to produce carbon dioxide, sulfur dioxide and water. After cooling to room temperature, the residual gas mixture was passed through aqueous potassium hydroxide.

What is the final volume of the resultant gas at room temperature and pressure after passing through the alkali?

- A 25 cm^3 B 35 cm^3 C 50 cm^3 D 75 cm^3

- 5 Paramagnetism is a relatively weak magnetic force produced by unpaired electrons. It is only detectable in the presence of external magnetic field. The strength of the force is directly proportional to the number of unpaired electrons in an atom or ion.

Which atom has the greatest paramagnetism?

- A** magnesium **B** silicon **C** phosphorus **D** sulfur

- 6 Which species has the biggest bond angle?

- A** CO_3^{2-} **B** SF_6 **C** NH_4^+ **D** OCl_2

- 7 Which of the following shows the correct response for the presence (indicated by ✓) or absence (indicated by x) of hydrogen bonds in the substances?

	$\text{NH}_3(\text{l})$	$\text{NH}_4\text{Cl}(\text{s})$	$\text{CH}_3\text{CONH}_2(\text{l})$
A	✓	✓	✓
B	✓	x	✓
C	x	✓	x
D	✓	x	x

- 8 Which types of bonding are present in lithium aluminum hydride, LiAlH_4 ?

- 1 co-ordinate (dative covalent)
- 2 ionic
- 3 metallic

- A** 1 only **B** 1 and 2 only **C** 2 and 3 only **D** 1, 2 and 3

- 9 A 25 cm³ solution containing of 0.20 mol dm⁻³ Ti⁺ required 25 cm³ of 0.10 mol dm⁻³ acidified KMnO₄ to oxidise it to Ti³⁺ in solution.

What is the oxidation state of the manganese in the reduced form?

- A +2 B +3 C +4 D +7

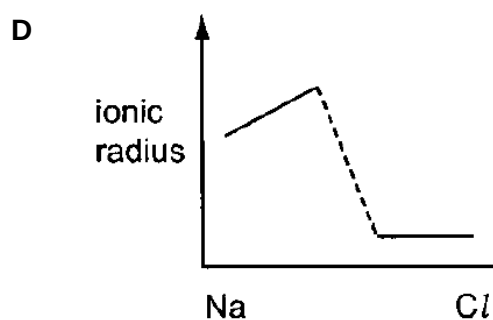
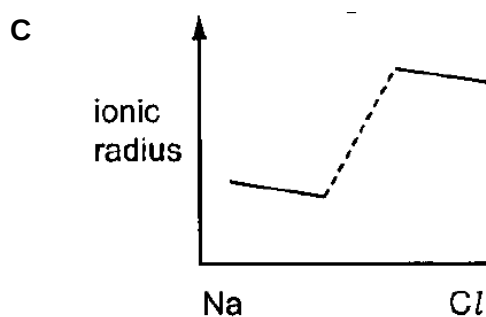
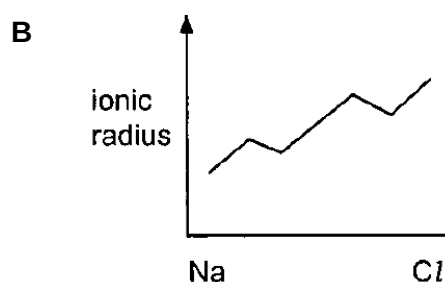
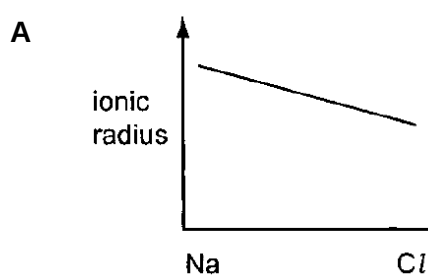
- 10 Sodium, silicon, sulfur and chlorine are elements in Period 3 of the Periodic Table.

Which of the following statements about these elements are true?

- 1 Sodium oxide has a giant lattice of ions.
- 2 Oxides of sulfur dissolve in water to form a solution that turns litmus red.
- 3 The chlorides of silicon have a higher melting point than that of oxides of silicon.
- 4 Sulfur exists as a yellowish gas as it has lower melting point than chlorine.

- A 1 and 2 only
B 1, 2 and 3 only
C 2 and 4 only
D 1, 3 and 4 only

- 11 Which diagram represents the change in ionic radius of the ions of the element across the third period (Na to Cl)?



- 12 The dissociation constant, K_w , for the ionisation of water, $2\text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-$, at different temperatures is given below.

temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$
0	1.15×10^{-15}
25	1.00×10^{-14}
50	5.50×10^{-14}

What can be deduced from this information?

- A $[\text{H}^+]$ and $[\text{OH}^-]$ are equal at 25 °C only.
- B The equilibrium lies more to the left as temperature increases.
- C The $[\text{H}^+]$ increases while the $[\text{OH}^-]$ decreases as temperature increases.
- D The pH of water at 50 °C is 6.6.
- 13 A solution was made by mixing 0.002 mol of $\text{HNO}_3(\text{aq})$ and 0.003 mol of $\text{Ba}(\text{OH})_2(\text{aq})$. Water was added until the volume of the solution was 1 dm^3 .

What is the pH of the solution?

- A 2.7 B 11.0 C 11.4 D 11.6
- 14 Which set of solutions of equal volume when mixed will give an acidic buffer?
- 1 0.05 mol dm^{-3} HCl and 0.10 mol dm^{-3} $\text{CH}_3\text{CO}_2\text{Na}$
 - 2 0.10 mol dm^{-3} $\text{CH}_3\text{CO}_2\text{H}$ and 0.10 mol dm^{-3} NaOH
 - 3 0.10 mol dm^{-3} H_2SO_4 and 0.05 mol dm^{-3} $\text{Ca}(\text{OH})_2$
- A 1 only
- B 1 and 2 only
- C 2 and 3 only
- D All

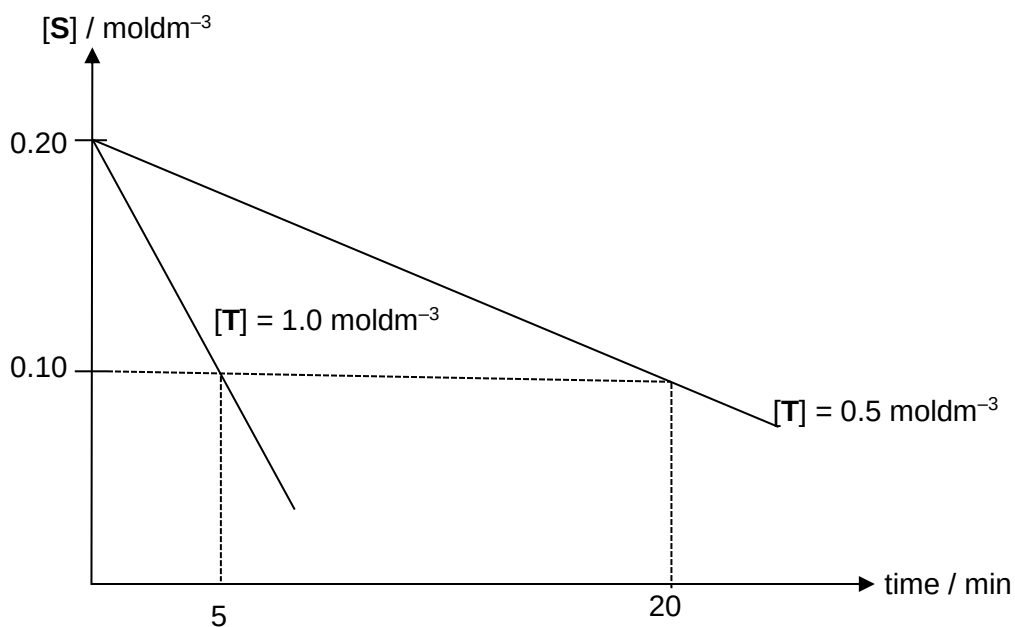
- 15 Xenon enhanced CT scanning is an easy and non-invasive method in which patients inhale the stable (^{131}Xe) gas while CT images are made. The CT scan is used in the study of pulmonary function and organ blood flow.

Xenon-131 is a radioactive isotope and it has a half-life of 11.8 days.

Given that radioactive decay is a first-order reaction, what fraction of the isotope has reacted after 59 days?

- A 0.0313 B 0.125 C 0.937 D 0.969

- 16 To monitor the kinetics of a reaction between **S** and **T**, two experiments were conducted to monitor the concentration of **S** over time with different concentrations of **T**. The graph below shows the results obtained.



What is the rate equation for the reaction between **S** and **T**?

- A rate = $k[\text{S}]$
 B rate = $k[\text{T}]^2$
 C rate = $k[\text{T}][\text{S}]$
 D rate = $k[\text{T}]$

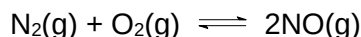
- 17 The rate of a reaction is measured in $\text{mol dm}^{-3} \text{ min}^{-1}$.

Which row is correct?

	Overall order of reaction	Units of the rate constant
A	zero	$\text{mol}^{-1} \text{ dm}^{-3} \text{ min}^{-1}$
B	first	s^{-1}
C	second	$\text{mol}^{-1} \text{ dm}^3 \text{ min}^{-1}$
D	third	$\text{mol}^{-3} \text{ dm}^9 \text{ min}^{-1}$

- 18 A sample of nitrogen monoxide gas is maintained at a constant temperature of 500 K.

A dynamic equilibrium is established with nitrogen, oxygen and nitrogen monoxide all present.



Which statement describes this situation?

- A At equilibrium, there are more NO gas present than N_2 and O_2 gas.
- B When pressure of reaction is reduced, more NO gas is produced.
- C Adding more NO gas after equilibrium has been established will shift position of equilibrium to the right.
- D At equilibrium, NO is formed at the same rate as it decomposes.
- 19 Given that the following reversible reaction take place at T °C,



Which of the following factor affects the K_c value?

- A Decrease in temperature.
- B Addition of suitable catalyst.
- C Increase in the concentration of X.
- D Addition of water

20 For which of the following reaction is the enthalpy change of reaction *endothermic*?

- 1 $\text{H}_2\text{SO}_4(\text{aq}) + 2 \text{NaOH}(\text{aq}) \rightarrow \text{Na}_2\text{SO}_4(\text{aq}) + 2 \text{H}_2\text{O}(\text{l})$
- 2 $\text{Cl}_2(\text{g}) \rightarrow 2 \text{Cl}(\text{g})$
- 3 $\text{Mg}(\text{g}) \rightarrow \text{Mg}^+(\text{g}) + \text{e}^-$
- 4 $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$

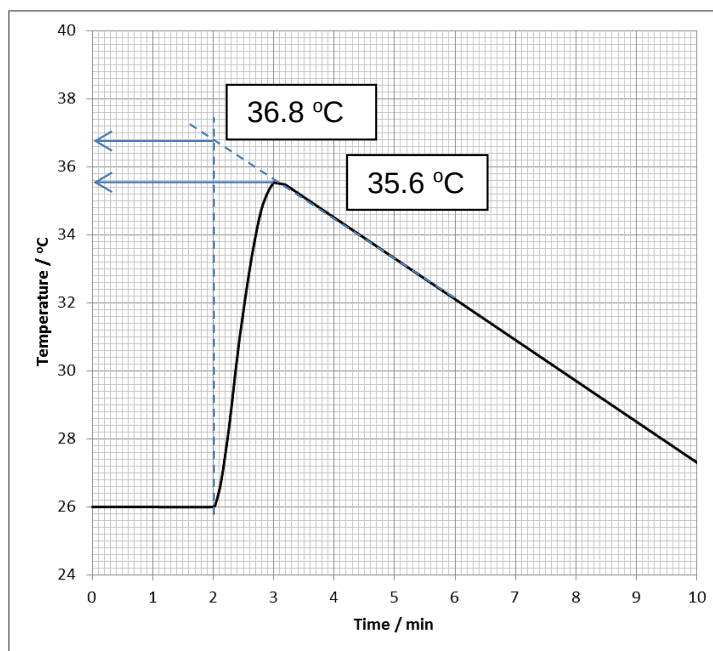
- A** 1 and 4 only
- B** 2 and 3 only
- C** 3 and 4 only
- D** 2, 3 and 4 only

21 Which compound has the most negative lattice energy?

- A** Rubidium fluoride
- B** Rubidium oxide
- C** Potassium fluoride
- D** Potassium oxide

- 22** A student conducted an experiment to determine the enthalpy change of solution of MgSO_4 in water. 1 mol of solid MgSO_4 was added to 100 g of water at 2 min. The changes in the temperature of the solution was monitored over a period of time.

Using the data obtained from the experiment, the following graph was obtained.

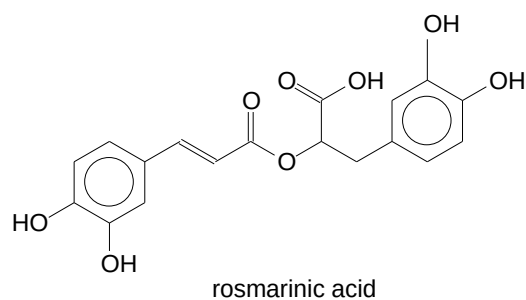


What is the enthalpy change of solution of MgSO_4 based on the data obtained from the experiment?

The specific heat capacity of water is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

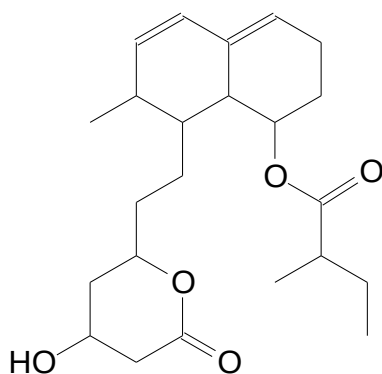
- A** $100 \times 4.18 \times (36.8 - 26.0)$
- B** $100 \times 4.18 \times (26.0 - 36.8)$
- C** $100 \times 4.18 \times (35.6 - 26.0)$
- D** $100 \times 4.18 \times (26.0 - 35.6)$

- 23 Rosmarinic acid occurs in culinary herbs such as rosemary, sage and thyme.



Which of the functional group is **not** present in rosmarinic acid?

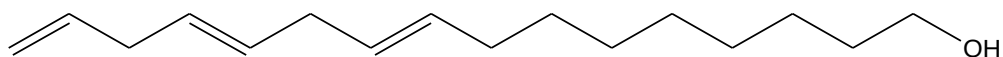
- | | |
|---|--|
| <p>A ester</p> <p>C carboxylic acid</p> | <p>B alcohol</p> <p>D alkene</p> |
|---|--|
- 24 Mevastatin belongs to the statin group of drugs that are used to lower cholesterol levels in humans.



What is the molecular formula of the compound?

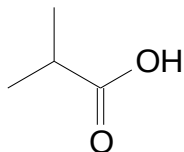
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|----------------------------|----------------------------|----------------------------|----------------------------|
| A $C_{23}H_{34}O_5$ | B $C_{23}H_{31}O_5$ | C $C_{22}H_{27}O_5$ | D $C_{22}H_{23}O_5$ |
|----------------------------|----------------------------|----------------------------|----------------------------|

- 25 How many cis–trans isomers are present in this molecule?



- A 2 B 3 C 4 D 8

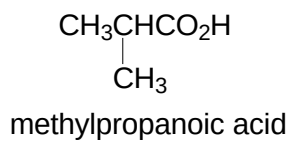
- 26 The carboxylic acid shown below can be produced by oxidation of an alcohol.



Which alcohol can be oxidised to produce this carboxylic acid when heated under reflux in the presence of an oxidising agent?

- A 2–methylpropan–2–ol
 B 2–methylpropan–1–ol
 C 2–methylbutan–1–ol
 D butan–1–ol

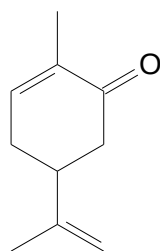
- 27 Ethyl propanoate is refluxed with aqueous sodium hydroxide. The alcohol produced is then reacted with methylpropanoic acid to make a second ester.



What is the structural formula of this second ester?

- A** $\text{CH}_3\text{—CH}_2\text{—O—}\overset{\text{O}}{\parallel}\text{C—CH—CH}_3$
 CH_3
- B** $\text{CH}_3\text{—}\overset{\text{O}}{\parallel}\text{C—O—CH}_2\text{—CH—CH}_3$
 CH_3
- C** $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—O—}\overset{\text{O}}{\parallel}\text{C—CH—CH}_3$
 CH_3
- D** $\text{CH}_3\text{—CH}_2\text{—}\overset{\text{O}}{\parallel}\text{C—O—CH}_2\text{—CH—CH}_3$
 CH_3

- 28 Carvone is responsible for the odour of spearmint.

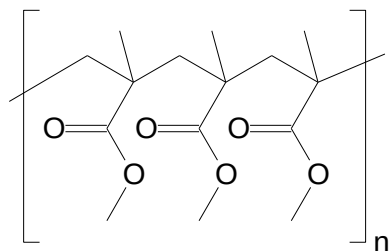


carvone

Which reagent and conditions, when added to carvone, would give observable changes?

- A Hot acidified potassium dichromate
 - B Aqueous bromine
 - C Warm dilute NaOH
 - D Ethanoic acid with concentrated H_2SO_4 , heat
- 29 Which of the following statements are true for thermoplastic polymers but not true for thermoset polymers?
- 1 The polymer burns upon heating.
 - 2 The polymer can be recycled.
 - 3 There is little or no cross-linkages between polymer chains.
- A 1 only
 - B 1 and 2 only
 - C 2 and 3 only
 - D 3 only

- 30 Poly(methyl methacrylate) is used to make hard contact lenses. Part of its polymer chain is shown.



Which statement about poly(methyl methacrylate) is correct?

- A A small molecule of water is lost when polymerisation took place.
- B Its monomer includes a dicarboxylic acid and a diol.
- C It can undergo alkaline hydrolysis.
- D It can form hydrogen bonds between two polymer chains.

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